

EXAM 7 – SPRING 2013

4. (2 points)

Given the following information:

<u>Cumulative Paid Loss (\$000)</u>				
<u>Accident Year</u>	<u>12</u>	<u>24</u>	<u>36</u>	<u>48</u>
2009	5,751	10,640	11,491	12,181
2010	5,528	9,287	10,680	
2011	4,120	7,004		
2012	5,304			

Calculated Ultimate Loss (\$000)

<u>Accident Year</u>	<u>Bornhuetter-Ferguson Ultimate</u>	<u>Benktander Ultimate</u>
2009	12,181	12,181
2010	11,246	11,316
2011	8,428	8,204
2012	10,403	10,609

a. (1.5 points)

Calculate the 24-month-to-ultimate cumulative development factor that would result in the ultimate loss estimates shown above.

b. (0.5 point)

For accident year 2011, suppose that the Bornhuetter-Ferguson method is performed over multiple iterations. Deduce the ultimate loss estimate that will be produced as the number of iterations approaches infinity.

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Question 4 Sample Answer

Solution 1

- a) $R_{GB} = q_k \times U_{BF}$
 $(8204 - 7004) = (1 - 1/LDF_{24-ULT}) \times 8428$
 $1200/8428 = 1 - 1/LDF_{24-ULT}$
 $LDF = 1.1660$
- b) As you iterate on the BF method to infinity, the result will approach the chain ladder estimate.

Solution 2

- a) AY 2011
 $U_{BF} = 8428 = 7004 + q_i \times U_0$
 $U_{BT} = 8204 = 7004 + q_i \times 8428 \rightarrow q_i = .1424$
 $p_i = \% \text{ Reported} = 1 - q_i = .8576$
 $LDF_{24-ULT} = 1/.8576 = \boxed{1.166}$
- b) $U_{BF} = U_{CL} \times (1 - q_i) + U_0 \times q_i \rightarrow 1^{st} \text{ iteration}$
 $U_{BT} = U_{CL} \times (1 - q_i^2) + U_0 \times q_i^2 \rightarrow 2^{nd} \text{ iteration}$
 $3^{rd} \text{ iteration: } U_{CL} \times (1 - q_i) + q_i [U_{CL} \times (1 - q_i^2) + U_0 \times q_i^2]$
 $U_{CL} \times (1 - q_i) + q_i \times U_{CL} \times (1 - q_i^2) + U_0 \times q_i^3$
 $U_{CL} \times (1 - q_i) [1 + q_i \times (1 - q_i)] + U_0 \times q_i^3$
 $U_{CL} \times (1 - q_i) [1 + q_i - q_i^2] + U_0 \times q_i^3$
 $\dots$
 $U_{CL} \times (1 - q_i^3) + U_0 \times q_i^3$

As the number of iterations approaches infinity

$$\rightarrow U_{CL} (1 - q_i^\infty) + U_0 \times q_i^\infty$$

where $\lim_{t \rightarrow \infty} q_i^t = 0$

thus the reserve estimate becomes $U_{CL} (1 - 0) + U_0 \times 0 = U_{CL}$

$$U_{CL} = 7004 \times LDF_{24-ULT} = 7004 \times 1.166 = \boxed{8167}$$

Examiner Comment

This question was generally answered well by candidates.

Part a)

The majority of candidates received full credit for part a). Errors, if any, were generally due to calculation/arithmetic errors. The critical component of the solution was the relationship between the Benktander and Bornhuetter Ferguson method.

Part b)

The question in part b) did not explicitly state that candidates needed to estimate the value, so full credit was given to candidates that mentioned chain-ladder ultimate without explicitly calculating the value of the chain-ladder ultimate.

A few candidates used a cumulative development factor inconsistent with their calculated cumulative development factor in part a). Partial credit was given to candidates with this response.

Some candidates derived the estimate showing iterations to infinity. These candidates received full credit if their derivation was correct.

Partial credit was given to candidates that calculated the value of the chain-ladder estimate, without explicitly stating that it is the chain-ladder estimate, or showing a derivation.